

PhoneInfoga – Technical & OSINT Research Overview

1. What is PhoneInfoga?

PhoneInfoga is an open-source OSINT (Open Source Intelligence) framework focused on phone numbers. It does not hack a phone or provide real-time tracking. Instead, it helps an investigator collect and correlate publicly available information about a given number.

Originally written in Python and later rewritten in Go, PhoneInfoga is now a cross-platform command-line and web tool. It can run locally, in Docker, or as part of other software via its REST API or Go modules. Typical outputs include number validity, country, carrier, line type, and online footprints such as spam reports, VoIP provider clues and appearances in public directories or search results.

2. Architecture & Technology Stack

- **Language & implementation** – Current stable versions are written in Go. Older, archived versions were Python-based. The tool is distributed as pre-built binaries for Linux, macOS and Windows, and also as a Docker image.
- **Deployment options** – It can be run directly as a CLI binary, via Docker or Docker Compose, or installed via package managers such as Homebrew on macOS. A built-in web server exposes a graphical interface and a REST API.
- **Modularity** – PhoneInfoga embeds multiple "scanners". Each scanner is responsible for a specific kind of lookup (basic numbering plan data, OSINT search, reputation checks, etc.). Scanners can use local logic, public web services or authenticated third-party APIs.
- **Input format** – Numbers must be supplied in international formats (E.164 or standard international notation). Internally, PhoneInfoga uses libraries that understand international numbering plans to normalise and validate input before scanning.

3. Core Features

PhoneInfoga focuses on information gathering, not exploitation. Key capabilities:

- **Basic intelligence**
 - Validates whether a number is syntactically possible for its country.
 - Returns country, region/area, carrier and line type (mobile, fixed line, VoIP, etc.).
- **OSINT footprinting**
 - Queries public phone books, search engines and reputation/spam-report sites.
 - Looks for indications that the number is disposable, virtual or tied to known VoIP services.
 - Searches for social-media and web footprints where publicly visible.
- **Reputation & risk hints**
 - Aggregates results from crowd-sourced spam lists and complaint forums.
 - Helps flag potential robocallers, scams or throwaway numbers.

- **Automation & integration**

- Browser-based GUI for quick manual checks.
- REST API endpoints for programmatic use in internal tools.
- Go modules for developers who want to integrate PhoneInfoga's scanners into their own software.

- **Batch work**

- Can process lists of numbers sequentially, useful during leak/breach investigations or fraud analysis.

4. Data Sources & Scanner Types

PhoneInfoga's power comes from how it orchestrates multiple lookups rather than from any secret database. Different scanners perform tasks such as:

- **Numbering plan & carrier lookup** – Uses libraries and public numbering plan data to infer country, region, carrier and line type.
- **Search-engine reconnaissance** – Builds targeted queries to see where the number appears on public websites, forums, paste sites or cached pages.
- **Public phone books & directories** – Checks online directories where available for name or address information.
- **Reputation / spam databases** – Queries public or API-based spam/abuse reporting services to see whether the number has a history of complaints.
- **OSINT APIs** – Optional scanners can call third-party APIs (commercial or free tiers) for enhanced data such as carrier info, reachability or risk scores. These require the user to supply their own API keys.

Every scanner has its own strengths and limits. Some work globally, others are country-specific. If a particular directory has strong privacy protection or no public API, PhoneInfoga simply cannot see into it.

5. Typical Workflow (Legitimate Use)

A typical, lawful workflow for an analyst might look like this:

- 1) Normalise the number – Convert the input to E.164 / international format (+CountryCodeNumber).
- 2) Basic scan – Run a basic scan to validate the number and retrieve country, carrier and line type.
- 3) OSINT scan – Run OSINT/reputation scans to see where the number appears online and whether it shows up in spam reports or known VoIP services.
- 4) Correlate manually – Cross-check results with internal data, case context and other OSINT sources such as breach corpora, open corporate records or email intelligence.
- 5) Document findings – Record what is confirmed versus speculative. PhoneInfoga's own documentation stresses that results are not guaranteed to be relevant, current or accurate and should never be the only source of truth.

6. Legitimate Use Cases

Examples of ethical, legitimate use for PhoneInfoga include:

- Fraud and abuse investigations – Enriching suspicious phone numbers seen in scam reports, phishing campaigns or account-takeover attempts.
- Incident response – Checking new numbers that appear in logs, alerts or phishing SMS messages.
- Threat intelligence and red teaming – Mapping infrastructure for social-engineering simulations or adversary profiling, within agreed rules of engagement.
- Training and education – Teaching OSINT techniques in controlled environments such as labs, capture-the-flag exercises or security courses.
- Personal security – Security professionals investigating persistent spam callers or verifying unknown numbers, while staying within local law and platform terms of service.

7. Limitations and "Anti-Features"

The maintainer explicitly documents what PhoneInfoga does NOT do:

- It does not claim to provide verified or fully accurate data; results are best-effort hints from public sources.
- It does not track a phone or its owner in real time.
- It does not provide precise GPS location.
- It does not exploit phones or help break into accounts.

Accuracy is constrained by the underlying data sources, rate limits, privacy controls and number reassignment. Investigators should treat PhoneInfoga as one signal among many, never as a standalone attribution tool.

8. Legal & Ethical Considerations

Because phone numbers are personal data in many jurisdictions, PhoneInfoga usage is constrained by privacy and data-protection law:

- A lawful basis (such as legitimate interest or contractual necessity) is needed in many regions before processing personal data for investigations.
- Using the tool for harassment, stalking, doxxing or unrelated background checks can violate criminal law, civil law and platform terms of service.
- Requests hit third-party websites and APIs; users must respect each provider's terms and rate limits.
- Logs, screenshots and exports from PhoneInfoga should be handled as sensitive data with defined retention, access control and secure disposal.

Any deployment in a company or agency should be reviewed by legal counsel and integrated into formal OSINT and privacy policies.

9. Security & Privacy Risks for Operators

Operating PhoneInfoga introduces some practical risks:

- API key exposure if configuration files are mishandled.

- Logging of investigated numbers and queries, which must be protected.
- Potential correlation of large volumes of lookups by external services, revealing investigative focus.
- Misconfigured Docker or web deployments that expose the interface beyond the intended team.

These can be mitigated with careful secrets management, restricted access to the interface and logs, network controls and use of hardened, monitored environments.

10. Comparison with Alternatives

Tool / Approach	Pros	Cons	Typical Role	Best For
PhoneInfoga (self-hosted OSINT tool)	Open source, locally controlled, extensible scanners, integrates into security workflows.	Requires technical setup; depends on external sources; results not guaranteed accurate; must be used lawfully.	OSINT enrichment and triage for investigators comfortable with CLI/web tools.	Security teams, researchers and trained OSINT analysts.
Consumer caller-ID / spam-blocking apps	Very easy to use; large crowd-sourced datasets; automatic caller ID and spam warnings.	Centralised commercial database; privacy trade-offs; limited transparency about sources and accuracy.	Everyday spam-call detection for end users.	Individuals wanting quick call screening, not deep investigations.
Manual OSINT without PhoneInfoga	Full control over every query and data source; can be tailored per case and jurisdiction.	Slower and more labour-intensive; higher chance of missing low-signal traces without automation.	Custom investigative workflows using browsers, search engines and scripting.	High-stakes cases where each step is carefully curated and documented.

11. Key Reference Types

This summary is based on:

- The official PhoneInfoga GitHub repository and documentation (project description, features, installation and scanner design).
- Public blog posts and tutorials showing how PhoneInfoga fits into OSINT workflows.
- General OSINT, privacy and data-protection guidance relevant to handling phone numbers as personal data.

For formal reports, you can cite the GitHub project page, the official documentation site and any jurisdiction-specific legal references you rely on.